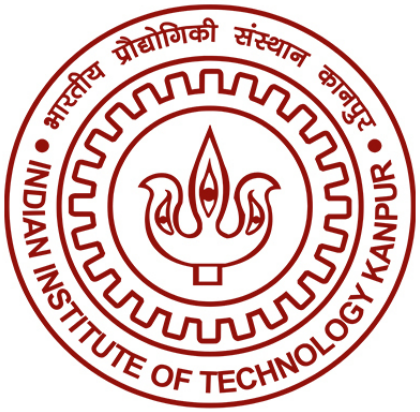



## SPA 617 - RADIO ASTRONOMY LAB

# RF transmission, reflection, standing waves

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**Lab Report**

# 1 Aim

To understand signal propagation, reflection and standing wave formation in a coaxial transmission line at radio frequencies and to determine important cable parameters experimentally.

# 2 Objectives

1. To determine the phase velocity and velocity factor of a signal in a coaxial cable, by measuring the frequency spacing between the consecutive minima produced the reflection caused in an open ended cable.
2. To study the reflection of the signal from an open cable and a cable terminated with a load $50\,\Omega$.
3. To estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched with a $50\,\Omega$ termination, indicating that its characteristic impedance is close to $50\,\Omega$.
4. To calculate the inductance and capacitor per unit length of the coaxial cable using the measured phase velocity and the approximate characteristic impedance of $50\,\Omega$.
5. To measure the insertion loss and attenuation per unit length of the coaxial cable.
6. To measure the length of an unknown length cable. (Bonus!)

# 3 Theory

In RF (radio frequency) applications, understanding transmission line behavior is essential for optimizing signal integrity, reducing losses, and ensuring efficient power delivery.

Transmission lines, such as coaxial cables, propagate electromagnetic signals and can experience reflections if the line's impedance mismatches with the load. The incident and reflected waves interfere with each other, producing maxima and minima in the measured signal amplitude. By varying the frequency, the phase of the reflected wave changes, causing a periodic variation in amplitude.

For an open-ended cable, the frequency difference between two consecutive maxima or minima can be used to determine the phase velocity:

$$v_p = 2l\Delta f$$

where $l$ is the cable length and $\Delta f$ is the frequency spacing.

The velocity factor is given by,

$$VF = \frac{v_p}{c}.$$

The voltage reflection coefficient is,

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0},$$

where $Z_L$ is the load impedance and $Z_0$ is the characteristic impedance of the cable. For an open circuit, $|\Gamma| = 1$, while for a matched load $Z_L = Z_0$, $\Gamma = 0$.

The inductance and capacitance per unit length of a low-loss transmission line are related to the characteristic impedance and phase velocity by

$$L' = \frac{Z_0}{v_p}, \qquad C' = \frac{1}{Z_0 v_p}.$$

The cable also introduces attenuation as the signal propagates. The insertion loss is determined by comparing the measured power with the cable to the reference power without the cable. The attenuation per unit length is then obtained by dividing the insertion loss by the cable length.

# 4 Equipments

- RF signal generator
- Digital storage oscilloscope(DSO)
- Spectrum analyser
- RF Tee connector
- Coaxial cable
- Short reference coaxial cable
- $50\,\Omega$ termination
- Cable of unknown length (Bonus!)

# 5 Experimental setup

The RF signal generator is connected to a Tee connector. One arm of Tee connector is connected to DSO, while the other arm is connected to the coaxial cable. The end of the coaxial cable is kept open. The frequency of the signal generator is varied, and the signal amplitude on the DSO is observed. Frequencies corresponding to two consecutive minima are noted as $f_1$ and $f_2$. The frequency difference is calculated as,

$$\Delta f = f_2 - f_1.$$

# 6 Observations and Results

First we measured the length of the cable as 92.5 cm.

After connecting all the cables with the connector, we observed the consecutive minimas using the Digital storage Oscilloscope (DSO).

Table 1: Frequency vs Voltage

| **Frequency(MHz)** | **Voltage(mV)** |
|---|---|
| 60 | 0.00 |
| 40 | 160 |
| 45 | 120 |
| 47 | 103 |
| 49 | 87 |
| 50 | 80 |
| 51 | 70 |
| 53 | 50 |
| 55 | 29 |
| 57 | 15 |
| 59 | 25 |
| 60 | 35 |
| 61 | 45 |
| 63 | 67 |
| 65 | 90 |
| 70 | 135 |
| 75 | 163 |
| 80 | 189 |
| 85 | 201 |
| 90 | 206 |
| 100 | 217 |
| 105 | 217 |
| 110 | 223 |
| 115 | 220 |
| 125 | 222 |
| 130 | 217 |
| 135 | 211 |
| 140 | 205 |
| 145 | 190 |
| 150 | 175 |
| 155 | 155 |
| 160 | 123 |
| 162 | 110 |
| 164 | 95 |
| 165 | 86 |
| 166 | 77 |
| 168 | 59 |
| 170 | 38 |
| 172 | 21 |
| 174 | 26 |
| 175 | 35 |
| 176 | 45 |
| 180 | 86 |
| 185 | 125 |
| 190 | 148 |
| 195 | 163 |
| 200 | 173 |

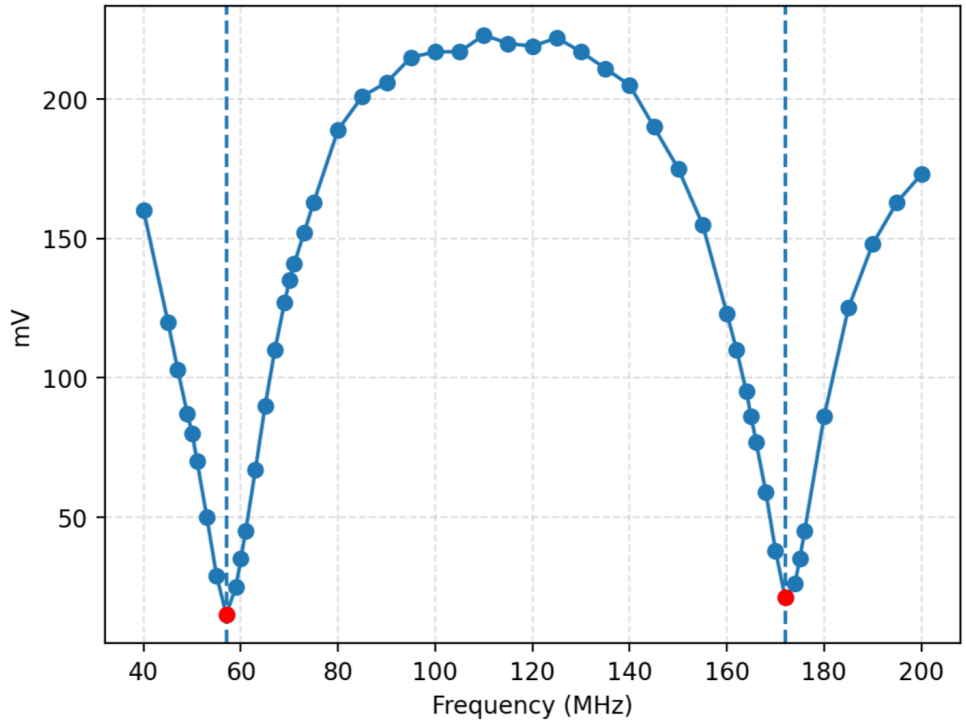



Figure 1: Red dots show consecutive minimas

The frequency difference between those minimas is, $\Delta f = f_2 - f_1 = (174 - 58)$ MHz $= 116$ MHz. After finding the frequency difference, we used that to find phase velocity,

$$v_p = 2\mathrm{l}\Delta f.$$

$$v_p = 2 \times 92.5 \times 10^{-2} \times 116 \times 10^6 \text{ m/s} = 2.13 \times 10^8 \text{ m/s}.$$

Next we calculated the velocity factor,

$$\mathrm{VF} = \frac{\mathrm{v_p}}{\mathrm{c}}.$$

$$\mathrm{VF} = \frac{2.13 \times 10^8}{3 \times 10^8} = 0.71.$$

For the characteristic impedance, we connected Tee connector to an open circuit, then reconnected with a 50 Ω. Retaining the same frequency as before to carefully determine the $V_{max,open}$ and $V_{min,open}$ for calculating the reflection coefficient. The same thing we have done with the 50 Ω termination load to note the difference in reflection coefficient.

Reflection coefficient for open circuit,

$$|\Gamma| = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} = \frac{1.7 - 0.048}{1.7 + 0.048} = 0.95.$$

Reflection coefficient for 50 Ω load,

$$|\Gamma| = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} = \frac{1.12 - 0.93}{1.12 + 0.93} = 0.09.$$

Table 2: Reflection coefficient for open and 50 Ω load

| **Termination** | $\mathbf{V}_{\max}$ (V) | $\mathbf{V}_{\min}$ (V) | $\lvert\mathbf{\Gamma}\rvert$ |
|---|---|---|---|
| Open circuit | 1.7 | 0.048 | 95% |
| 50 Ω load | 1.12 | 0.93 | 9% |

This large reduction in reflection coefficient proves that the cable is approximately matched with the 50 Ω termination load.

The characteristic impedance as calculated taking both signs of Γ,

$$Z_0 = Z_l \times \frac{1-\Gamma}{1+\Gamma}.$$

$$Z_0 = 50 \times \frac{1-0.09}{1+0.09} = 41.7\ \Omega.$$

and

$$Z_0 = Z_L \times \frac{1+\Gamma}{1-\Gamma} = 50 \times \frac{1+0.09}{1-0.09} = \ 59.8\Omega.$$

Inductance per unit length,

$$\mathrm{L}' = \frac{\mathrm{Z_0}}{\mathrm{v_p}} = \frac{41.7}{2.13 \times 10^8}\ \mathrm{H/m} = 19.59 \times 10^{-8}\ \mathrm{H/m}.$$

Capacitance per unit length,

$$\mathrm{C}' = \frac{1}{\mathrm{Z_0} \times \mathrm{v_p}} = \frac{1}{41.7 \times 2.13 \times 10^8}\ \mathrm{F/m} = 0.011 \times 10^{-8}\ \mathrm{F/m}.$$

For the cable attenuation per unit length the spectrum analyser has been used to record reference power $P_{ref}$ and $P_{cable}$. Calculated attenuation per unit length,

$$\alpha = \frac{\mathrm{InsertionLoss}}{\mathrm{Length}} = \frac{\mathrm{P_{ref}} - \mathrm{P_{cable}}}{\mathrm{l}} = \frac{0-(-0.65)}{0.925}\ \mathrm{dB/m} = 0.7\ \mathrm{dB/m}.$$

Table 3: Attenuation of the cable

| **Frequency(MHz)** | $P_{ref}(dBm)$ | $P_{cable}(dBm)$ | $\alpha\ (dB/m)$ |
|---|---|---|---|
| 60 | 0.00 | -0.65 | 0.7 |

Table 4: Experimental results for coaxial cable

| **Quantity** | **Symbol** | **Measured/Calculated Value** |
|---|---|---|
| Cable length (m) | $\ell$ | 92.5 cm |
| Frequency spacing (Hz) | $\Delta f$ | 116 MHz |
| Phase velocity (m/s) | $v_p$ | $2.13 \times 10^8$ m/s |
| Velocity factor | $v_p/c$ | 0.71 |
| Approximate characteristic impedance ($\Omega$) | $Z_0$ | 41.7 $\Omega$ |
| Inductance per unit length (H/m) | $L'$ | $19.59 \times 10^{-8}$ H/m |
| Capacitance per unit length (F/m) | $C'$ | $0.011 \times 10^{-8}$ F/m |
| Attenuation (dB/m) | $\alpha$ | 0.7 dB/m |

## 6.1 Bonus part

Table 5: For calculation of unknown length of cable

| **Minima** | **Frequency(MHz)** | **Voltage(V)** |
|---|---|---|
| 1st | 52 | 120 |
| 2nd | 87 | 170 |

Frequency difference $\Delta f = (87 - 52)$ MHz $= 35$ MHz.
Length of the unknown cable,

$$l = \frac{v_p}{2 \times \Delta f} = \frac{c \times \text{VF}}{2 \times \Delta f} = \frac{3 \times 10^8 \times 0.71}{2 \times 35 \times 10^6} \text{ m} = 3.04 \text{ m}.$$

Error calculation for the unknown length:

$$\% \text{ Error} = \left| \frac{3.06 - 3.04}{3.04} \right| \times 100\% \tag{1}$$

$$= 0.65\%. \tag{2}$$

# 7 Conclusion

The experiment provided comprehensive insight into the behavior of transmission lines at radio frequencies, successfully determining the phase velocity, attenuation, characteristic impedance, and length of an unknown cable. When load is not matched to the characteristic impedance of the cable, a part of the signal is reflected back towards the source. By changing the frequency and observing the consecutive minimas, the propagation velocity and other parameters of coaxial cable like velocity can be determined experimentally. Observations revealed that phase velocity and velocity factor correlate closely with the dielectric properties of the coaxial cable.

By changing the terminating end of the wire, the reflected wave also changes.

In the case of short circuit, the reflection coefficient is $-1$. It shows that reflected wave is of the same magnitude but with opposite phase as compared to the open end circuit. In case of 50 $\Omega$ load the reflection coefficient becomes 0, and hence no reflected wave for interference.

Then we accurately estimated the unknown cable length with the help of velocity factor and frequency difference. We calculated the cable length as 3.04 meter which was very close to its actual length 3.06 meter.